Fiscal Year:	FY 2026	**Task Last Updated:**	FY 02/27/2026
PI Name:	Britten, Richard Ph.D.		
Project Title:	Changes in the Neuroproteome Associated with HZE-Induced Impairment of Cognition		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	02/27/2014	**End Date:**	10/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: Element change from SR to HFBP per Human Research Roadmap information dtd July 2019 (Ed., 1/6/2020) NOTE: End date changed to 10/31/2025 per S. Mack-Phillips / NASA JSC (Ed., 6/2/25) NOTE: Extended to 6/30/2025 per S. Mack-Phillips / NASA JSC (Ed., 5/16/24) NOTE: Extended to 6/30/2024 per NSSC information (Ed., 8/3/23) NOTE: Extended to 4/30/2023 per NSSC information (Ed., 12/30/2020) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 3/12/19) NOTE: Extended to 12/31/2018 per S. Monk/SR/LaRC (Ed., 1/11/18)		

Key Personnel Changes/Previous PI:	April 2025 update per the PI: Due to the late stage of the project, Co-Investigators are no longer with the project. These have been removed from the list of Co-Investigators for this reporting period: Douglas Wallace, Ashley Blackwell, Biswajit Maharathi (Ed., 4/17/25). December 2022 report: Dr. Dorela Shuboni-Mulligan of Eastern Virginia Medical School has been added to the project to determine if the SR-induced cognitive decrements are related to changes in the retina, specifically the loss of intrinsically photosensitive retinal ganglion cells (ipRGCs). December 2022 report: Dr. Biswajit Maharathi from the Illinois Brain Analytics Institute (University of Illinois Chicago) has been added to the project to assist in applying AI-assisted learning algorithms to determine neural network cohesiveness. December 2021 report: Dr. Jessica Burket at Christopher Newport University has been added to the project to conduct the social interaction studies. Dr Ashley Blackwell at Eastern Virginia Medical School has been added to the project to conduct the string pulling and neural network cohesiveness studies. December 2019 report: Dr. Douglas Wallace at Northern Illinois University (NIU) is now a CoInvestigator. Dec 2016: Drs. Semmes and Dutta were removed from the project; proteomic analysis is now being conducted at UTMB (University of Texas Medical Branch) as contract work.
COI Name (Institution):	
Grant/Contract No.:	NNX14AE73G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	One of NASA's biggest concerns about the long-term health of astronauts who go on deep space missions is the impact that High Z, High Energy (HZE) particles have on brain function (neurocognition). Data from our laboratory and others suggests that there is significant impairment of certain neurocognitive tasks (spatial memory and Executive function-attentional set shifting) following exposure to low HZE doses. The goal of this application is to determine the Threshold dose for the induction of HZE-induced spatial memory impairments (HISMI) or Attentional Set Shifting Impairments (HIASSI) following exposure to 56Fe, 48Ti, and 28Si particles. The proposed studies will also identify the changes in the proteome of the brain (neuroproteome) of rats that differ in their susceptibility to HISMI and HIASSI, which will provide further insight into the factors that lead to HISMI/HIASSI and perhaps more importantly, that prevent its emergence. Our underlying hypothesis is that HISMI and HIASSI arise as the direct result of HZE-induced changes in the neuroproteome. We also hypothesize that exposure to HZE species that have different track structures will result in different mechanisms of HZE-induced cognitive impairment (HICI). Collectively, these studies will give some insight into the underlying cause for HISMI and HIASSI. Our studies will thus address CNS (Central Nervous System) Gaps 1, 2, and 6, [Ed. note February 2022: Human Research Program risks and gaps have changed since this project was initiated--see Human Research Roadmap for updated gaps: https://] and we shall specifically focus on the following aims: Aim 1. Determine the Threshold dose for the induction of HISMI and HIASSI following exposure to 56Fe, 48Ti, and 28Si particles when delivered as a single dose. Aim 2. Identify changes in the neuroproteome that are associated with susceptibility or resistance to developing HISMI and HIASSI following exposure to 56Fe particles. Aim 3. Determine the mechanism of HISMI and HIASSI induced by HZE particles of differing LET (linear energy transfer). In Aim 1, socially mature (~6 month old) male Wistar rats will be irradiated with 56Fe, 48Ti, and 28Si particles (with incident energies of 600 MeV/nucleon). Rats will receive whole body HZE irradiation (< 15 cGy), and HISMI and HIASSI will be assessed at 3 months post irradiation. In Aim 2 and 3, the composition of the neuroproteome (hippocampus and selected regions of the prefrontal cortex) of irradiated rats that have “normal” cognitive performance or have developed HISMI or HIASSI will be established using an unbiased proteomic profiling approach. We shall use a label free differential protein profiling workflow on the Q-Exactive Orbitrap mass spectrometer. These studies will give considerable insight into the underlying cause for HZE-induced neurocognitive failure. The proposed studies will continue to define the minimum dose of HZE particles that will induce HISMI and HIASSI. Moreover, our studies will provide considerable insight into the underlying mechanism of HICI, and will identify prognostic biomarkers that could be translated to human studies to monitor the emergence of HICI. These studies may also help to develop appropriate countermeasures and help identify sensitive individuals, so that NASA’s medical staff can implement appropriate countermeasures to protect these at risk individuals. Supplemental studies (in December 2019 report) This study will provide information on the robustness of single-exposure experiments to predict the impact of repeated episodic radiation exposures (such as will be encountered on the mission to Mars) on neurocognition. This study will test the hypothesis that episodic SR exposure will result in more severe neurocognitive deficits than single, or multiple daily SR doses. In addition, this study will be a robust (akin to a pHase III clinical trial) concurrent validation of the effect of a single dose of 10 cGy simplified 6-ion GCRSim versus a single dose of 10 cGy 250 MeV/n He ions on ATSET/UCFlex performance using the same batch of rats, laboratory personal, transport and environmental conditions. This study will utilize both male and female rats, and two different radiation regimens incorporating 4He ions and the 6-ion GCRsim beam. Executive function performance (ATSET) will be assessed after a single exposure (He or GCRsim) and after a second exposure (~6 months later) to the 6-ion GCRsim beam. To maximize the amount of data obtained from these expensive studies, where possible (dependent upon volunteers in the Britten lab) the impact of these radiations on sensorimotor (string pulling activity), social interaction and switch task performance will also be established.
Rationale for HRP Directed Research:	

Research Impact/Earth Benefits:	Our studies have established that that SR exposure can impair key cognitive and motor domains critical for astronaut performance, including attention, flexibility, decision-making, and fine motor coordination. The observed sex- and individual-dependent variability emphasizes the need for personalized risk models and countermeasure strategies. Proteomic and behavioral biomarkers identified in these studies provide a foundation for precision medicine approaches to predict and mitigate SR-induced cognitive decline.
Task Progress:	Our studies have: • Defined radiation doses and biological contexts that produce operationally relevant cognitive and motor impairments. • Identified molecular signatures and neurophysiological pathways underlying space radiation (SR) vulnerability and resilience. • Demonstrated that repeated or combined exposures may exacerbate deficits through cumulative or priming effects. • Provided translational assays and analytic frameworks to support NASA's risk modeling, countermeasure development, and astronaut health monitoring during exploration-class missions. Key Findings and Outcomes from the entire project: 1. LET, Dose Rate, and Spectrum Effects: Attentional Set Shifting (ATSET) performance is impaired by SR doses as low as 1 cGy. Across helium, silicon, titanium, iron, and mixed-ion exposures, no clear LET- or ion-specific dependence was observed in the pattern of ATSET impairment. Both acute and protracted low-dose-rate exposures induced comparable deficits, suggesting that the biological effect of SR on cognition is driven more by cumulative dose and target tissue sensitivity than by delivery mode or ion composition. 2. Sex-Dependent and Individual Variability: Male and female rats displayed distinct patterns of SR-induced impairment. Males typically showed associative memory deficits, whereas females exhibited slower processing and perseverative responding during reversal and set-shifting stages. Across all cohorts, 20–36% of animals showed overt impairments, reflecting substantial inter-individual variability in cognitive resilience, an observation directly relevant to astronaut crew heterogeneity. 3. Loss of Practice Effects and Learning Efficiency: Both male and female rats exposed to SR exhibited reduced or reversed practice effects, failing to improve performance on repeated task exposure. This loss of learning efficiency implies a diminished capacity for skill acquisition and cognitive adaptation under stress, critical operational concerns for long-duration missions. 4. Domain-General Cognitive Deficits: SR exposure produced broad impairments across multiple cognitive domains. In addition to constrained flexibility (ATSET), SR reduced performance in tasks requiring unconstrained flexibility (UCFlex), interference control (ARMIT), and task switching, with distinct sex-dependent manifestations. These converging results demonstrate radiation-induced disruption of frontostriatal and hippocampal-prefrontal circuits supporting attention, working memory, and decision-making. However, these performance decrements were dissociable at the individual level. 5. Decision-Making and Risk Propensity: In risk-taking predisposition (RTP) tasks, SR exposure altered cost–benefit evaluation strategies. He-exposed rats showed increased preference for risky choices, while GCRsim exposure produced sex-specific effects: males exhibited global reward insensitivity, and females favored high-risk, low-reward options. These results implicate dopaminergic and orbitofrontal network dysfunction and suggest that SR exposure can bias decision-making under uncertainty. 6. Sensorimotor Impairment: SR-exposed rats demonstrated rapid and persistent deficits in fine motor coordination using the string-pulling task. Increased approach times, grasp errors, and bilateral coordination deficits were observed as early as 72 hours post-exposure. These findings indicate that sensorimotor systems are highly sensitive to even low SR doses, posing a potential risk to manual dexterity and EVA task performance. 7. Neuroproteomic and Pathological Correlates: Proteomic profiling of hippocampal and medial prefrontal cortex tissue revealed downregulation of proteins involved in oxidative phosphorylation, mitochondrial metabolism, and synaptic signaling. Impaired proteostasis and energy regulation were strongly associated with cognitive decline. Complementary histopathology identified SR-induced microglial activation, aberrant neurogenesis, and ectopic granule cell formation, indicating persistent neuroinflammation and circuit-level remodeling. 8. Radioadaptive and "Two-Hit" Vulnerability: Re-irradiation studies revealed that prior SR exposure alters subsequent cognitive outcomes. Rats that were cognitively intact after an initial 10 cGy exposure exhibited new impairments after a second exposure, particularly during attentional set-shifting (IDR/EDS) stages, characterized by slowed processing speed. These results support a two-hit vulnerability model, wherein SR primes corticostriatal and frontoparietal networks for collapse under subsequent challenges (e.g., additional SR or sleep loss).
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Britten RA, Tamgue EN, Alvarado-Arriaga PA, Fesshaye AS, Sanford LD. "A two-hit model of executive dysfunction: simulated Galactic Cosmic Radiation primes latent deficits revealed by sleep fragmentation. " Life (Basel). 2025 Nov 6;15(11):1717. https://doi.org/10.3390/life15111717 ; PubMed PMID: 41302141; PubMed Central PMCID: PMC12653965 , Nov-2025
Articles in Peer-reviewed Journals	Britten RA, Tamgue EN, Arriaga PA, Li N, Phuyal S. "Exposure to low (10 cGy) doses of simulated space radiation impairs reward-guided decision making in both male and female rats." Life Sci Space Res. 2026 Mar 22. Online ahead of print. https://doi.org/10.1016/j.lssr.2026.03.008 , Mar-2026